Chile's Vera Rubin telescope

It has world's largest digital camera

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High in the Andes Mountains of Chile, something is happening. Perched on the peak of Cerro Pachón at an altitude of 2,682 meters, the Vera C. Rubin Observatory is set to change the way we explore the night sky. Built for the most ambitious astronomy project, the Legacy Survey of Space and Time (LSST) is not just a powerful telescope. It's a game-changer for science.

Here are five really incredible facts that not only show what the telescope can do, but also how groundbreaking this facility really is.

Interstellar Visitor Spotted

In the early hours of July 1, 2025, Chile's ATLAS telescope at El Sauce Observatory spotted 3I/ATLAS, a comet from another star system, and the third interstellar object to visit us streaking near Jupiter's orbit. Likely up to 20 kilometers wide, it's also possibly the largest. Unlike 'Oumuamua, which was seen too late to study, 3I/ATLAS was caught early, giving us months to investigate it. It will be closest to the Sun in October, and nearest to Earth at a safe 150 million miles.

Read more on: https://tinyurl.com/3IatLsp

The largest digital camera

At the core of the Rubin Observatory is a technological giant: the largest digital camera in the history of astronomy. It weighs almost three tons and is about the size of a small car. With a 3.2-gigapixel resolution (that's 3,200 megapixels), this camera can capture incredibly sharp images. Its sensor is 5.5 feet wide and made up of 189 CCD detectors.

The camera's wide field of view, 3.5 degrees, or about seven times the size of the full moon, means it can photograph huge areas of the sky at once. In April 2025, it took its first official images, revealing breathtaking details of nebulae and distant galaxies.

An unbelievable amount of data

Every night the observatory is active, it collects 20 terabytes of data, about the same as taking 5 million smartphone photos every single night. Over the course of 10 years, it will gather around 500 petabytes, which is enough to store the entire content of Wikipedia a 1,000 times over.

This data will be used to build a map of the universe, including stars, galaxies, and objects in our solar system. The observatory has advanced systems to process and share this information in real time with scientists around the world.

Catch cosmic events in real time

The Rubin Observatory doesn't just take pictures, it watches the sky for changes. Its telescope will scan the entire southern sky every few nights and can spot things like supernovae, gamma-ray bursts, and other fast-changing events within 60 seconds of them appearing. d

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